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(57) Abstract:

ABSTRACT Disclosed herein is a context-aware user interface adaptation system (100) for virtual reality applications, comprising a camera unit (102), a user interface (104) integrated into a virtual reality device (106), a communication network (108), a processing unit (110) comprising an input module (112), a preprocessing module (114), a perception module (116) configured to produce a latent user-state vector via convolutional and recurrent encoding, a momentum module (118) configured to suppress adaptation during high-momentum flow states, a persona module (120) configured to detect cognitive behavioral modes and activate persona-specific policy branches, a policy module (122) executing a Proximal Policy Optimization agent, a conflict module (124) configured to impose conservative adaptation upon reward disagreement, an output module (126), and a database (128).